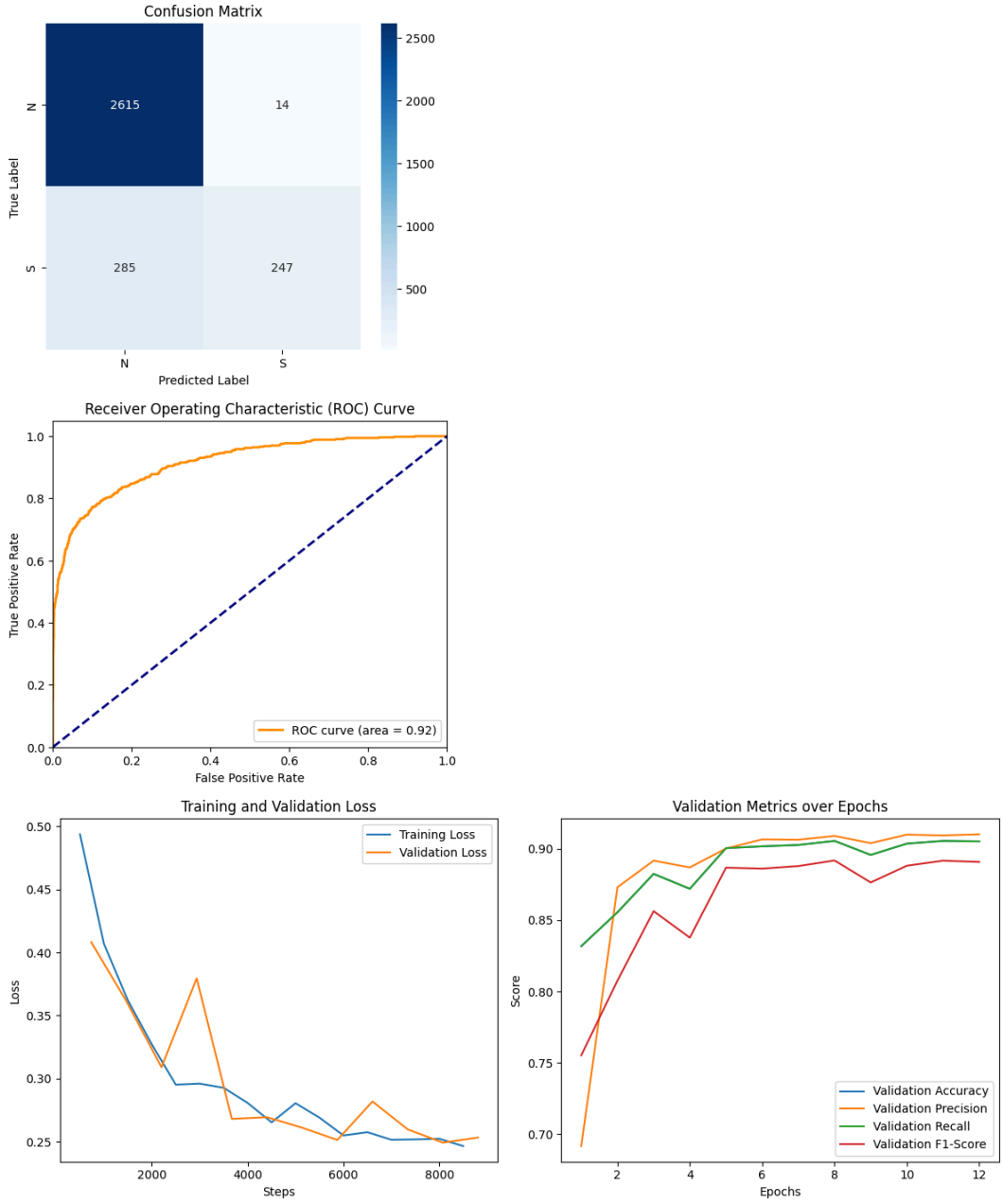
```
    plt.plot(epochs, eval_f1, label='Validation F1-Score')
    plt.title('Validation Metrics over Epochs')
    plt.xlabel('Epochs')
    plt.ylabel('Score')
    plt.legend()
else:
    plt.text(0.5, 0.5, 'No validation metrics to display', horizontalalignment='center', verticalalignment='center', transform=plt.gca().transAxes)
    plt.title('Validation Metrics over Epochs')

plt.tight_layout()
plt.show()

# Summarize implementation of linear projection layer and evaluation results
summary = f"""
Implementation Details of Linear Projection Layer:
- A `linear_projection` layer (nn.Linear) was added after the mean pooling step in `CustomBertForSequenceClassificationWithMeanPooling`.
- It reduces the dimensionality of the pooled embeddings from `config.hidden_size` (1024 for bert-large-portuguese-cased) to `config.reduced_embedding_dim` (256).
- A GELU activation function is applied after the linear projection.
- The final classification head (`self.classifier`) then takes this 256-dimensional vector as input.
- This approach aims to reduce the number of parameters in the classification head, potentially speeding up training and reducing the risk of overfitting, especially when freezing most of

Evaluation Results:
{metrics}
"""
print(summary)
```

```
Getting predictions on the test set...
Evaluation Metrics: {'accuracy': 0.9054096804808605, 'precision': 0.9092364315574998, 'recall': 0.9054096804808605, 'f1': 0.8915651604448387}
```




```
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